

JAYANTH S

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AI/ML Research Engineer with a background in working with deep learning models, building LLM-based retrieval systems, and applying machine learning to wireless communication problems. Hands-on experience with end-to-end ML systems, retrieval pipelines, and experimentation workflows. Current interests include Machine Learning, Stochastic Optimization, Reinforcement Learning, and applied research.

EDUCATION

Indian Institute of Technology, Dharwad

2020 – 2022

M.S(Research) in Electrical Engineering, CGPA = 9.17/10

Dharwad, Karnataka

Thesis supervisor: Rajshekhar V Bhat

PES Institute of Technology, Bangalore South Campus

2015 – 2019

B.E. in Electronics and Communication Engineering, CGPA = 8.16/10

Bengaluru, Karnataka

EXPERIENCE

Nokia Standards

Mar 2025 -

6G AI/ML Research Specialist

Bengaluru, Karnataka

- Built **LLM-based retrieval workflows** for technical knowledge tasks such as summarization, question answering, information extraction, and cross-document comparison.
- Developed **RAG pipelines** using embeddings and vector search over structured and unstructured technical data.
- Contributed to **React-based frontend** and **Streamlit prototypes** for internal AI tools, including backend integration and feature updates.
- Designed multi-step workflows using **LangChain** and **LangGraph**, with prompt engineering for task routing, reasoning, and automation.
- Worked on **deep learning models** for signal processing and communication systems, including multi-antenna scenarios, performance evaluation, and benchmarking against conventional methods.
- Conducted large-scale **model experimentation and simulation studies**, including architecture updates, loss-function tuning, and robustness analysis under varied conditions.
- Migrated and validated a large-scale simulation and ML codebase (**approx. 10k LOC, 80 modules**) from **TensorFlow to PyTorch**, including porting integrated neural network components, debugging, verification, and simulation-based testing.
- Implemented reproducible **ML experimentation pipelines** using **MLflow, Flyte, and Argo Workflows** for experiment tracking, scheduling, and scalable execution.
- Documented and enabled **cross-framework model portability workflows** for deploying trained deep learning models across simulation environments, improving reuse and adoption within the team.

TCS Research

Feb 2023 - Mar 2025

Research Engineer

Bengaluru, Karnataka

- Won the **Depth Estimation using mmWave AI/ML Challenge - 2023** organized by **NIST (USA)** and **ITU**, with solution presentation in the final event [[Video Presentation](#)].
- Implemented **OFDM** and **OTFS** communication systems on **USRP B210 SDRs** for real-time wireless prototyping and over-the-air transmission experiments.
- Demonstrated image/data transmission using **OFDM** over SDR platforms with a **Reconfigurable Intelligent Surface (RIS)**-assisted wireless link.
- Developed **cascaded channel estimation algorithms** for RIS-aided mmWave communication systems, resulting in publication at **IEEE ICC Workshop 2024**.

Linköping University

Oct 2022 - Dec 2022

Research Assistant, Department of Computer and Information Science

Linköping, Sweden

- Worked on stochastic control methods for optimizing **semantic communication efficiency**, **Age of Information**, and cost-aware scheduling policies with Assoc. Prof. **Nikolaos Pappas**.
- Research outcomes led to a **WiOpt 2023 publication**, with subsequent extensions resulting in an additional conference publication as co-author.

TCS Research

May 2022 - Aug 2022

Research Intern

Bengaluru, Karnataka

- Implemented **DDPG, TD3, and SAC** reinforcement learning algorithms for hybrid beamforming in single-user MIMO communication systems.

PROJECTS AND PAPER IMPLEMENTATIONS

nanoQwen – LLM Implementation from Scratch [\[code\]](#) 2026

- Implemented a Qwen-style decoder-only Transformer from scratch in **PyTorch** with **RoPE**, **RMSNorm**, **Grouped-Query Attention (GQA)**, causal self-attention, and autoregressive text generation.
- Built end-to-end **pretraining** and **supervised fine-tuning (SFT)** pipelines with custom dataset preprocessing, mixed-precision training, checkpointing, configurable experiment workflows, and tokenizer integration.
- Developed modular training and inference infrastructure including **KV-cache** based decoding, **HuggingFace** parity validation, model evaluation utilities, and configurable scaling for transformer experimentation.

Policy Gradient Algorithms for Atari Games [\[code\]](#) 2022

- Applied **A2C**, **A3C**, **TRPO**, and **PPO**, based on the *stablebaselines3* and *ray rllib* implementations for Pong, Breakout and Space-Invaders atari games.

Implementation of the Paper - Model Free Training for End-to-End Communication Systems [\[code\]](#) 2022

- Implemented the model-based and model-free **auto-encoder** based end-to-end communication system for AWGN and Rayleigh Block Fading(RBF) channels as given in the paper: Fayçal Ait Aoudia and Jakob Hoydis, “Model-Free Training of End-to-End Communication Systems.”

TECHNICAL SKILLS

Languages: Python, C++, MATLAB

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Ray RLlib

GenAI Systems / Tooling: LangChain, LangGraph, Streamlit, Elasticsearch, Retrieval-Augmented Generation (RAG), Prompt Engineering

MLOps/ Dev Tools: Git, MLflow, Flyte, Argo Workflows

Relevant Coursework: Reinforcement Learning, Statistical Pattern Recognition, Convex Optimization, Probability and Stochastic Processes

PUBLICATIONS

1. **Jayanth S**, Vishnu Karthikeya Gorty, A Anil Kumar, Tapas Chakravarty, Arpan Pal, “RIBCE: RIS-BS virtual array based Channel Estimation for mmWave communication system,” **IEEE ICC Workshop, 2024**.
2. **Jayanth S**, Nikolaos Pappas, Rajshekhar V Bhat, “Distortion Minimization with Age of Information and Cost Constraints,” **WiOpt, 2023**.
3. **Jayanth S** and Rajshekhar V Bhat, “Age of Processed Information (AoPI) minimization with power constraint in fading multiple access channels,” **IEEE ICC 2022** and the extended version of the paper is published in **IEEE Transactions on Wireless Communications(IEEE-TWC)** journal.
4. Gagan G B, **Jayanth S** and Rajshekhar V Bhat, “Age of Information Minimization with Power and Distortion Constraints in Multiple Access Channels,” **WiOpt, 2021**.

PATENTS

- **Filed Patent:** Application No. **202421019098**, Method and System for Channel Estimation in Reconfigurable Intelligent Surface (RIS) Assisted Communication System, *TCS Research*.

CERTIFIED COURSES

- Deep Learning Specialization- Coursera
- Reinforcement Learning Specialization - Coursera